

Socioeconomic Vulnerability and Shooting Violence in New York City: A Census-Tract Analysis

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Abstract

Firearm violence remains a major public health and safety challenge in the United States, disproportionately affecting disadvantaged urban neighborhoods. This study examines the association between socioeconomic vulnerability and shooting violence across New York City census tracts by integrating 24,127 NYPD shooting records with 2024 American Community Survey data, 2022 CDC/ATSDR Social Vulnerability Index, and 2024 TIGER/Line census boundaries. After spatial aggregation and complete-case screening, the final dataset included 2,185 census tracts and 17 standardized predictors. Three complementary models - Multiple Linear, Negative Binomial, and a logistic model were evaluated. Households without vehicles, limited English, single-parent households, lower educational attainment, and disability were consistently associated with higher violence, while educational attainment was protective. The logistic model separates high-shooting tracts on held-out data with an accuracy of 0.799 and AUC of 0.89. The findings support place-based prevention and equitable resource allocation.

Keywords: firearm violence; social vulnerability; census tracts; regression analysis

Introduction

Problem Context

Firearm violence remains one of the most pressing public-health and public-safety challenges in the United States, producing fatal and nonfatal injuries as well as psychological trauma, economic disinvestment, reduced mobility, and weakened confidence in community institutions (Cook & Ludwig, 2006; Wintemute, 2015). Although New York City has experienced substantial declines in overall crime during the past three decades, shooting incidents remain disproportionately concentrated in a relatively small number of neighborhoods. This persistent clustering suggests that structural neighborhood conditions, rather than individual characteristics alone, play an important role in shaping firearm risk. The American Psychological Association concluded that no single psychological profile or warning sign can reliably identify who will commit firearm violence because violent behavior develops through interacting individual, family, peer, school, community, and sociocultural influences (American Psychological Association, 2013). Population-level research similarly connects firearm violence to social determinants of health, including economic deprivation, educational opportunity, housing conditions, and access to community resources (Kim, 2019). Together, these findings support shifting the analytical focus from identifying inherently dangerous individuals to examining how neighborhood conditions influence exposure to violence and a community's capacity for prevention.

We chose this topic because firearm violence remains a significant public-health and public-safety concern, yet its bur-

den is distributed unevenly across New York City neighborhoods. Citywide averages can obscure substantial differences among communities, particularly when shootings remain concentrated despite broader reductions in crime. The availability of detailed geocoded NYPD shooting records, census-tract socioeconomic indicators from the American Community Survey, and multidimensional measures from the CDC Social Vulnerability Index provides an opportunity to investigate whether structural neighborhood conditions help explain these disparities. To examine the relationship from multiple perspectives, this study models population-adjusted shooting rates, incident counts, and the probability that a tract will be classified as high-shooting using linear, count-based, and logistic regression approaches. Comparing results across these complementary models allows us to identify which socioeconomic and vulnerability measures consistently predict elevated firearm violence without relying on the assumptions of a single model. A clearer understanding of these relationships may support more equitable, place-based prevention strategies and resource allocation.

Literature Review

Social disorganization theory provides the principal framework for explaining why neighborhood context matters. In its classical formulation, concentrated poverty, residential instability, family disruption, and related structural pressures weaken the local networks and institutions through which residents maintain shared expectations and informal social control. Sampson and Groves (1989) found that structural disadvantage operated partly through weakened friendship net-

works, limited organizational participation, and reduced supervision of teenage peer groups. Bursik (1988) clarified that social disorganization should not be treated as a synonym for visible disorder or as a moral judgment about residents. Rather, it describes a community's reduced capacity to regulate behavior collectively when resources, stable relationships, and institutional connections are constrained. This distinction is central to the present study because census-tract indicators such as unemployment, poverty, housing burden, and single-parent households should be interpreted as markers of structural conditions, not as labels applied to the character of the people who live in those tracts.

The concept of collective efficacy extended social disorganization theory by specifying the social process that links structural conditions to violence. Collective efficacy combines mutual trust among neighbors with a willingness to intervene for the common good. Sampson et al. (1997) found that collective efficacy accounted for a substantial portion of the association between concentrated disadvantage and violent crime, indicating that structural deprivation matters partly because it constrains residents' ability to coordinate action. Morenoff et al. (2001) added a spatial dimension, showing that a neighborhood's violence level is influenced by conditions in surrounding areas as well as within its own boundaries. Reviews of neighborhood-effects research have consequently emphasized that structural disadvantage, social processes, residential selection, and spatial interdependence must be considered together (Kubrin & Weitzer, 2003; Sampson et al., 2002). This theoretical progression supports a census-tract analysis while also cautioning that tract boundaries are analytical approximations rather than self-contained social systems. The expected association between vulnerability and shootings is therefore not simply a compositional effect; it reflects how overlapping resource constraints can influence informal control, institutional capacity, and exposure across connected places.

A broad empirical literature supports the importance of structural disadvantage. Early comparative studies identified poverty, unemployment, income inequality, family disruption, and educational disadvantage as persistent macro-level correlates of homicide and violent crime (Land et al., 1990). Pratt and Cullen (2005)'s meta-analysis found that measures of concentrated disadvantage and social disorganization were among the most reliable macro-level predictors of crime, while McCall et al. (2010) showed that many of these structural relationships remained evident when the classic homicide-covariate literature was reassessed two decades later. The consistency of these findings does not imply that a single variable is sufficient. Poverty, employment, education, household structure, and housing conditions are correlated but theoretically distinct dimensions of vulnerability. Their effects may accumulate, interact, or become especially consequential after disadvantage reaches a high level. Krivo and Peterson (1996), for example, documented particularly elevated crime in extremely disadvantaged neighborhoods, suggesting

that severe concentration can produce conditions not captured by a simple linear interpretation of any one indicator.

Income inequality and access to social resources add another layer to this structural account. Cross-national research has linked greater inequality to higher violent-crime rates even after considering average income, suggesting that relative deprivation and unequal opportunity may matter alongside absolute poverty (Fajnzylber et al., 2002). Messner et al. (2004) similarly connected dimensions of social capital to homicide, emphasizing that networks of trust and civic engagement can shape a community's capacity to regulate conflict. Hipp (2007) showed that the internal distribution of race and class can affect crime rates in ways that are obscured by aggregate averages. These studies reinforce the value of analyzing several socioeconomic indicators simultaneously, but they also create a methodological challenge: the same structural process can be represented by multiple highly correlated variables. A coefficient may therefore change magnitude or direction after adjustment, not because the underlying social process has reversed, but because the model is estimating the residual association that remains after related dimensions of disadvantage have been held constant.

Education provides a distinct but closely connected pathway. Additional schooling may reduce criminal involvement by increasing legitimate economic opportunities, raising the opportunity cost of illegal activity, changing peer networks, and reducing unstructured time during high-risk years. Quasi-experimental studies have found that increases in schooling reduce criminal participation and arrest outcomes (Lochner & Moretti, 2004; Machin et al., 2011). At the neighborhood level, higher educational attainment can also indicate stronger labor-market access, greater institutional resources, and a larger capacity to navigate public and private systems. Conversely, a high share of adults without a high-school diploma may signal long-standing exclusion from opportunity. Education should not be interpreted independently of income and employment, however. Its relationship with violence is embedded in a broader system of labor demand, residential segregation, family resources, and institutional investment. For this reason, the present analysis treats educational attainment as both a substantive predictor and a potential source of multicollinearity with other measures of socioeconomic vulnerability.

Household composition is often included in neighborhood models because family structure can reflect both resource constraints and the availability of informal supervision. High rates of single-parent households are frequently associated with violence in macro-level research, but the pathway is not that single parenthood directly produces crime. Rather, a household with one adult may face greater time, income, childcare, and transportation constraints, particularly when those pressures are concentrated across a neighborhood and combined with limited institutional support. Sampson and Groves (1989) treated family disruption as one of several structural conditions that can reduce community supervision,

while Land et al. (1990) and McCall et al. (2010) identified family-related measures among recurring covariates of homicide. The most defensible interpretation is therefore contextual: household composition may indicate the cumulative demands placed on residents and local organizations. It should be analyzed alongside employment, education, poverty, housing, and access to transportation, and any large adjusted coefficient should be examined for collinearity rather than interpreted as the effect of family structure in isolation.

Research on racial and ethnic disparities further demonstrates why structural interpretation is essential. Youth and adults in disadvantaged communities experience unequal exposure to firearm violence, and these disparities often remain visible even when overall citywide violence declines (Henry et al., 2024; Kravitz-Wirtz et al., 2022). Yet racial composition should not be treated as a direct causal explanation for shootings. Hipp (2007) showed that race and class distributions are intertwined with neighborhood inequality, while Lee et al. (2024) found that the relationship between residential racial segregation and youth firearm aggression operated indirectly through neighborhood disadvantage and exposure to violence. These findings support an interpretation centered on segregation, concentrated disadvantage, and unequal exposure rather than race itself. They also anticipate the possibility of suppression effects in multivariable regression: once poverty, education, employment, housing, and family structure are included, the remaining coefficient for a demographic-composition variable may represent a narrow residual contrast and may even reverse its bivariate sign. Such coefficients require careful contextual interpretation and should never be presented as isolated causal effects.

Neighborhood disadvantage and violence also appear to influence each other over time. Hipp (2010) described a reciprocal relationship in which structural characteristics affect crime while crime, in turn, changes residential composition, investment, and neighborhood stability. Semenza et al. (2023) extended this dynamic to community health, finding that concentrated disadvantage predicted subsequent gun-violence exposure and that violence was associated with worsening physical health, mental health, and health-related behavior. This reciprocal process helps explain why high-violence areas can remain persistently disadvantaged even when a cross-sectional model identifies only contemporaneous associations. Shootings can discourage commercial activity, alter daily routines, increase stress, and weaken attachment to place; these consequences may then reduce the very resources that support collective efficacy. A tract-level regression cannot establish the direction of this cycle, but it can identify where the cycle is most visible and which structural indicators consistently distinguish higher- and lower-burden neighborhoods.

The geography of firearm violence provides additional evidence that place matters. Spatial analyses have repeatedly shown that shootings are concentrated within a small fraction of urban space and that high-risk locations can remain stable over long periods. Braga et al. (2010) documented the con-

centration and persistence of gun violence at micro-places, while Weisburd (2015) described a broader law of crime concentration in which a relatively small number of street segments account for a substantial share of crime. Ceccato and Westman (2024) systematic review identified disadvantaged neighborhoods, transportation nodes, commercial corridors, and mixed land uses as recurring contexts for firearm-related violence. These patterns show that socioeconomic disadvantage is important but not sufficient by itself: violence often concentrates more tightly than poverty or unemployment, indicating that local opportunity structures, routine activities, social networks, and the built environment also shape risk. Census tracts are therefore useful for studying structural context, but they should be understood as a bridge between citywide inequality and the finer micro-places where incidents occur.

Spatial dependence also complicates conventional regression analysis. Baller et al. (2001) demonstrated that homicide rates exhibit spatial patterning that can alter the estimated effects of structural covariates when neighboring areas are treated as independent. Morenoff et al. (2001) similarly showed that violence and collective efficacy spill across neighborhood boundaries. More recent firearm-specific studies report that vulnerability in adjacent communities can contribute to shooting risk within the focal tract (Gill et al., 2026). These findings suggest that a tract may be affected by nearby commercial corridors, housing developments, transportation systems, and social networks even when those features fall outside its formal boundary. The current study does not claim to fully model every spatial process, but its use of fine-grained census tracts is preferable to broad borough- or city-level aggregation, and its interpretation should acknowledge that residual spatial clustering may remain after socioeconomic predictors are included.

Social-network research complements the geographic perspective by demonstrating that gun violence is not randomly distributed among people within high-risk places. Papachristos (2009) showed that patterns of gang homicide are embedded in durable social structures, while Papachristos et al. (2012) found that network position was associated with the risk of gunshot injury. Papachristos et al. (2015) likewise linked network exposure to homicide victimization. This work does not imply that violence spreads mechanically or that network membership determines behavior. Instead, it shows that exposure, retaliation, conflict, and victimization are structured through relationships that intersect with neighborhood conditions. Place-based and network-based explanations are therefore complementary: disadvantaged environments can increase exposure to risky settings and constrain protective resources, while social networks determine how risk moves among individuals within and across those environments. A tract-level model cannot observe these interpersonal mechanisms directly, but it can identify the structural contexts in which network-based risks are more likely to accumulate.

The built environment is another mechanism connecting structural disadvantage to violence. Vacant and poorly maintained properties can reduce guardianship, signal weak institutional responsiveness, and create spaces that facilitate illegal activity. Branas et al. (2012) found associations between vacant properties and neighborhood violence, while Kondo et al. (2018) reviewed evidence that greening vacant lots, improving public spaces, and other place-based environmental interventions can reduce violence. These findings are important because they demonstrate that neighborhood risk is modifiable and that prevention need not rely exclusively on policing or individual behavior change. Built-environment interventions may strengthen informal control by increasing legitimate use of space, visibility, and resident engagement. At the same time, their effects depend on broader socioeconomic conditions; physical improvements without sustained investment, employment opportunity, and community participation may not address the processes that reproduce vulnerability. The literature therefore favors integrated strategies that combine environmental change with social and economic supports.

The intervention literature also clarifies how place-based findings should be translated into policy. Concentration creates an opportunity to focus resources where harm is greatest, but it also creates a risk that high-burden communities will be subjected primarily to surveillance and enforcement. Cook and Ludwig (2006) argued for evidence-based gun policy, while Kondo et al. (2018) showed that environmental and public-health interventions can complement criminal-justice strategies. Research in New York City public housing further demonstrates that reforms such as body-worn cameras and revised surveillance practices must be evaluated in relation to community context and harm reduction, not only enforcement outputs (Braga et al., 2024). A responsible use of neighborhood prediction would therefore support investments such as trauma services, youth programming, street outreach, housing improvement, and environmental remediation rather than labeling a tract or its residents as dangerous. This ethical distinction is especially important when models contain race, language, disability, or household characteristics that can be misread outside their structural context. The goal is to identify unequal conditions and unmet needs, not to justify unequal treatment.

Contemporary public-health scholarship synthesizes these mechanisms under the broader concept of social and structural determinants of community firearm violence. Buggs et al. (2022) argued that firearm violence emerges from interacting systems involving economic inequality, housing, education, labor markets, structural racism, community resources, and exposure to prior violence. This perspective rejects explanations that isolate a single demographic variable and instead treats risk as multidimensional. It also distinguishes between identifying associations and assigning blame. A neighborhood with high vulnerability is not inherently violent; rather, its residents may face a greater concentration of conditions that increase exposure while reducing access to

prevention, trauma care, and institutional support. This framing aligns with the objective of the present study: to quantify neighborhood-level associations in a way that can guide equitable resource allocation without stigmatizing communities or implying that regression coefficients represent immutable causal traits.

The CDC/ATSDR Social Vulnerability Index provides a practical way to operationalize this multidimensional perspective. Originally developed for disaster management, the SVI summarizes a community's capacity to prepare for, respond to, and recover from hazardous events using indicators related to socioeconomic status, household characteristics, racial and ethnic minority status and language, and housing type and transportation (Flanagan et al., 2011). Although firearm violence differs from a natural disaster, both involve unequal exposure, unequal capacity to absorb harm, and unequal access to recovery resources. The SVI is therefore attractive because it captures overlapping vulnerabilities that may be missed by poverty or income alone. Its use also creates interpretive challenges. Composite themes and individual ACS indicators can measure similar underlying conditions, producing high correlations and unstable adjusted coefficients. Consequently, the value of the SVI may lie more in its joint predictive contribution and its ability to identify clusters of vulnerability than in a causal interpretation of any single component.

Empirical applications of the SVI increasingly support its relevance to firearm violence. Dirago et al. (2024) linked geocoded firearm assaults in Chicago to census-tract vulnerability and found that communities with greater social vulnerability experienced higher firearm burdens. Polcari et al. (2023) extended the approach across five U.S. cities, demonstrating that the relationship was not unique to one local context. McMillan et al. (2024) found that multidimensional social factors provided a broader account of urban gun violence than isolated indicators considered separately, and Gill et al. (2026) reported associations with both fatal and nonfatal shootings as well as effects involving neighboring communities. Taken together, these studies indicate that vulnerability is not a single pathway but a configuration of economic, household, demographic, housing, and transportation conditions. They also reveal a remaining question: whether SVI measures provide information beyond conventional ACS variables when both are entered into the same model and evaluated on held-out observations.

New York City offers a particularly useful setting for addressing that question. The city contains substantial variation in income, educational attainment, housing conditions, transportation access, residential density, and demographic composition within a single policy environment. Research comparing Philadelphia, New York, and Los Angeles found that poverty and unemployment explained only part of the spatial concentration of shootings during the COVID-19 period (MacDonald et al., 2022). Studies of New York City public-housing contexts further show that structural vulnerability shapes not only exposure to violence but also resi-

dents' experiences of surveillance and police reform (Braga et al., 2024). Earlier evaluations of order-maintenance policing also illustrate the difficulty of separating neighborhood change, police activity, and violence trends using aggregate observational data (Rosenfeld et al., 2007). These findings support a model that treats shootings as a public-health outcome embedded in neighborhood conditions while remaining cautious about causal claims and about interpreting enforcement activity as a direct measure of violence.

Measurement choices are central because different public-safety datasets capture different phenomena. Police stops reflect enforcement decisions, complaints reflect reporting and classification practices, homicide data capture only fatal events, and broad violent-crime categories combine offenses with different mechanisms. Geocoded shooting-incident records offer a narrower and more consistent outcome for studying firearm violence, although they still reflect the accuracy of administrative recording and geocoding. Aggregating incidents to census tracts permits linkage to ACS and SVI indicators and supports population-adjusted comparisons, but it also introduces the modifiable areal unit problem: associations may change when boundaries or geographic scales change. A rigorous interpretation should therefore treat the tract as a useful analytical unit rather than a natural boundary and should avoid assuming that a tract-level association applies to every individual resident.

Temporal variation presents a related challenge. Crime patterns can change with economic shocks, public-health emergencies, mobility, seasonal conditions, enforcement practices, and changes in the use of public space. MacDonald et al. (2022) documented a surge in shooting concentration during the COVID-19 period across several large cities, while Qiao et al. (2025) found that mobility and temperature contributed predictive information for New York City crime over time. Rosenfeld et al. (2007) also illustrated how difficult it is to attribute changes in homicide or robbery to a single policing strategy when broader neighborhood and city conditions are evolving simultaneously. These studies suggest that coefficients estimated from one period may not remain identical in another. A train-validation-test design cannot solve every temporal limitation when observations are cross-sectional, but held-out evaluation does reduce the risk of selecting a model solely because it fits the observed sample. The present study should therefore emphasize generalization within the available data and frame future longitudinal or spatial-temporal analysis as a necessary extension.

The distribution of shooting outcomes also has implications for model selection. Shooting counts are nonnegative integers, frequently contain many low-count observations, and often exhibit variance substantially greater than the mean. Osgood (2000) showed why Poisson-based regression is preferable to ordinary least squares for many aggregate crime counts, particularly when exposure differs across places. When the Poisson equidispersion assumption fails, negative binomial regression provides a more appropriate

count specification while preserving a population offset and an incidence-rate interpretation. A continuous shooting-rate model remains valuable because it offers accessible coefficient estimates and can be improved through transformations when rates are strongly skewed. Logistic regression addresses a different practical question by estimating the probability that a tract falls into a high-shooting category. No single model is universally superior; count, rate, and classification models represent complementary views of the same underlying pattern.

The comparison of multiple regression frameworks also addresses a tradeoff between prediction and explanation. Machine-learning approaches can capture nonlinear relationships and interactions, and recent New York City research has identified mobility and environmental conditions as useful predictors of urban crime (Qiao et al., 2025). However, highly flexible models may provide limited guidance about the direction and uncertainty of specific socioeconomic associations. Regression models retain policy value because coefficients, incidence-rate ratios, odds ratios, confidence intervals, and diagnostic tests can be interpreted directly. Their transparency does not eliminate model dependence, multicollinearity, or omitted-variable bias, but it makes those limitations visible. Evaluating linear, count-based, and logistic models under a common training, validation, and test framework can therefore provide both interpretability and evidence about generalization, while agreement across models strengthens confidence that the central vulnerability pattern is not an artifact of one response definition.

The existing literature has several limitations that the present study is designed to address. Regression-based studies often use broad geographic units, focus on homicide rather than all shooting incidents, or examine a narrow set of socioeconomic variables. Spatial hot-spot studies identify where violence concentrates but may offer limited explanation of why neighborhoods differ. SVI studies demonstrate that vulnerability and shootings co-vary, yet they do not always compare the index with detailed conventional socioeconomic measures or evaluate whether individual dimensions remain informative after adjustment. Studies also vary in their outcomes, time periods, geographic boundaries, and treatment of population exposure, making results difficult to compare. Finally, many analyses emphasize in-sample statistical significance rather than performance on held-out data. By combining geocoded NYPD shooting incidents, 2024 ACS socioeconomic characteristics, 2022 SVI indicators, and census-tract boundaries within a harmonized multi-model framework, the present study links structural explanation, spatial precision, and predictive evaluation more directly than any one of these approaches alone.

Prior Expectations

Taken together, the literature presents firearm violence as a preventable, spatially concentrated, and structurally conditioned public-health problem. Structural disadvantage may

weaken collective efficacy and institutional resources, while segregation, inequality, social networks, nearby places, and features of the built environment can concentrate or transmit violence across neighborhood boundaries. Repeated exposure to shootings may then further damage community health and stability, reinforcing existing disadvantage over time. Accordingly, the purpose of this study is not to claim that individual socioeconomic characteristics directly cause firearm violence or that high-risk neighborhoods are defined by their residents. Instead, the analysis evaluates whether a consistent place-based pattern emerges across multiple shooting outcomes and regression models, while maintaining a clear distinction between association and causation.

Based on this literature, we expect census tracts with greater economic and household disadvantage—including higher poverty and unemployment, lower educational attainment, housing and transportation vulnerability, family disruption, and greater social vulnerability—to exhibit higher shooting counts, higher population-adjusted shooting rates, and a greater likelihood of classification as high-risk. These characteristics are expected to operate as overlapping indicators of constrained opportunity and limited community capacity rather than as isolated causal mechanisms. We also expect SVI measures to provide predictive information beyond traditional ACS indicators, although multicollinearity may weaken individual estimates or produce adjusted sign reversals. Finally, model performance should depend on the outcome being examined: a count model that accommodates overdispersion should characterize shooting frequency more appropriately than a Poisson model, while logistic regression may provide the strongest discrimination of high-shooting tracts. These expectations are consistent with research showing that disadvantage interacts with community cohesion and structural conditions to sustain firearm violence over time (Henry et al., 2024; Kravitz-Wirtz et al., 2022), and they support the use of the findings to inform place-based prevention and equitable resource allocation.

Analysis

Data Preparation and Common Analytical Framework

We construct a tract-level analytical dataset for New York City by integrating four primary sources: (1) geocoded NYPD Shooting Incident data (New York City Police Department, 2025), (2) socioeconomic indicators from the 2024 American Community Survey (ACS) 5-year estimates (U.S. Census Bureau, 2024a), (3) multidimensional vulnerability measures from the 2022 CDC/ATSDR Social Vulnerability Index (SVI) (Agency for Toxic Substances and Disease Registry, 2022), and (4) 2024 TIGER/Line tract boundaries for spatial aggregation (U.S. Census Bureau, 2024b). The NYPD dataset contains latitude and longitude coordinates in reversed columns; we corrected these coordinates before the spatial join to ensure that shooting incidents were assigned to the appropriate census tracts.

The outcome variable for the count and linear models is `shooting_count`, the number of unique shooting incidents per tract, which we use to derive the population-adjusted rate `shooting_rate_per_1000` for the linear model. For the logistic model, we define a binary outcome `high_shooting_tract`, flagging tracts whose shooting rate exceeds the citywide median. An upper-quartile threshold was considered during early planning, but the median was adopted for the final specification because it yields a more balanced and stable classification task, avoiding the reduced effective sample size and less reliable probability estimates that a 25/75 class split would introduce. To model rates while accounting for varying population sizes, the count model includes `total_population` as an exposure offset.

After restricting the data to the five NYC boroughs and removing tracts with zero population or missing predictors, the final analytical sample comprises 2,185 census tracts. We split the data into training (60%), validation (20%), and test (20%) sets using a fixed random seed to ensure reproducibility. The training set is used for model fitting, the validation set for variable selection and hyperparameter tuning, and the test set is reserved exclusively for final evaluation. All 17 candidate socioeconomic and SVI predictors are standardized using means and standard deviations calculated from the training set only, ensuring that no information from the validation or test sets leaks into preprocessing.

Multiple Linear Regression Model

Methods We fit two specifications on the cleaned training data and compared them on the validation set on the rate scale: a model of the raw rate (Spec A) and a model of $\log(1 + \text{rate})$, back-transformed with `expm1` (Spec B). Spec A attained a marginally lower validation RMSE (3.13 vs 3.27) – a metric dominated by a few extreme tracts under squared-error loss – but Spec B produced a lower MAE (1.79 vs 1.94), a substantially better fit on its own scale (adjusted R^2 0.71 vs 0.60), and better-behaved residuals. We selected the log specification because our goal is interpretable, assumption-checked inference rather than minimizing error on a handful of outliers.

Our analytic sample is 2,185 NYC census tracts, which we obtain after restricting to the five boroughs, converting the CDC –999 suppression codes to missing, and dropping incomplete cases. We partition the data 60/20/20 into training, validation, and test sets with a fixed seed (6414) – the same procedure the count and logistic models use, so all three share the seeded protocol – and we standardize all 17 candidate predictors on training-set means and standard deviations only, so that each coefficient measures the effect of a one-standard-deviation increase and is comparable across predictors and across our three models. We then screen for influential observations on the training data alone: from the full 17-predictor fit, 16 tracts have externally studentized residuals beyond |3|, which we keep as extreme but genuine neighborhoods, and 68 exceed the Cook’s distance $4/n$ threshold, which we remove, leaving 1,243 training tracts. We never alter the validation or

test sets.

We begin variable selection from all 17 candidates. A variance-inflation screen confirms every predictor is below the conventional threshold of 10, and we then apply backward elimination – dropping the least significant term whenever any p -value exceeds 0.05, and tracking validation RMSE at each step – until we reach the project’s floor of 10 predictors. Validation RMSE rises only slightly across the path (3.27 to 3.31, about 1.3%), and a partial F -test of the reduced model against the full 17-predictor model is not significant ($F = 1.70$, $p = 0.11$), confirming the seven dropped predictors add nothing jointly. We refit the selected model on the combined cleaned-training and validation data ($n = 1,680$). The residual diagnostics reject constant variance and normality, so we base all coefficient inference on HC3 heteroscedasticity-robust standard errors, and we report predictive performance once, on the held-out test set ($n = 437$), on both the modeled log scale and the back-transformed rate scale.

Results Our linear model gives a clear, interpretable picture of how neighborhood socioeconomic vulnerability shapes the continuous shooting rate across New York City census tracts.

Structural determinants of the shooting rate. Under HC3 robust standard errors, we find all ten retained predictors significant at the 5% level, and the model is highly significant overall ($F(10, 1669) = 364$, $p < 0.001$; in-sample adjusted $R^2 = 0.68$ on the modeled log scale). We standardize the predictors, so each coefficient is directly comparable – the change in $\log(1 + \text{rate})$ per one-standard-deviation increase. The largest effect we estimate is the SVI minority-status ranking (+0.38 per SD), essentially matched by the share of households without a vehicle (+0.31) and, with the opposite sign, the limited-English share (-0.30). Bachelor’s attainment (-0.19), population density (-0.13), and the poverty rate (+0.12) form the next tier, followed by single-parent households (+0.09), crowded housing (-0.05), the unemployment rate (+0.04), and median age (-0.04). This pattern is what social disorganization theory predicts: concentrated vulnerability, household instability, unemployment, and constrained transportation access track higher rates, while educational attainment tracks lower rates. Figure 1 translates these standardized effects into predicted shooting rates: a tract two standard deviations above the mean on the SVI ranking is predicted at about 4.5 shootings per 1,000 – roughly triple the citywide-average tract (≈ 1.6) – while a comparably high limited-English share pulls the predicted rate toward zero.

The role of poverty. Unlike the count model, where poverty was collinear with other disadvantage measures and dropped from the stepwise specification, poverty survives our selection and is positive and significant (+0.12 per SD, $p < 0.001$). Holding education, vulnerability composition, density, and the remaining predictors fixed, a one-standard-deviation increase in the tract poverty rate raises $\log(1 + \text{rate})$ by 0.12.

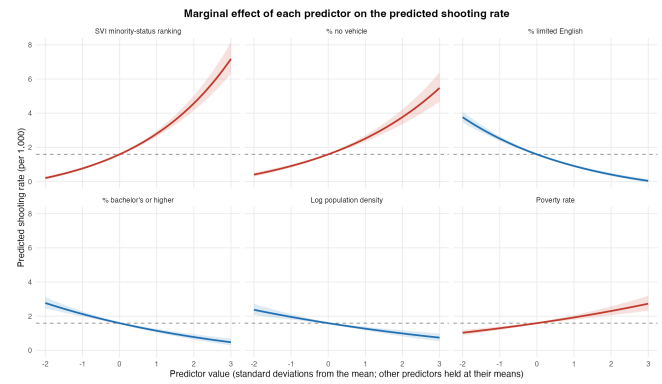


Figure 1: Marginal effect of the six strongest predictors on the predicted shooting rate. Each predictor is swept from -2 to $+3$ SD with the others at their means; bands are 95% CIs, the dashed line is the citywide-average tract (≈ 1.6), red = risk, blue = protective.

Our linear model therefore answers the project’s poverty question directly, within the multivariable specification rather than through a separate bivariate check.

Social vulnerability and out-of-sample performance. The SVI minority-status ranking is the single largest effect and stays dominant alongside the conventional ACS indicators (poverty, unemployment, education), so the composite vulnerability ranking carries information beyond the standard socioeconomic variables. This holds out of sample: on the held-out test set our model explains 58% of the variation on the log scale ($R^2 = 0.58$) and 35% on the back-transformed rate scale ($R^2 = 0.35$; RMSE 3.06, MAE 1.6 shootings per 1,000), close to its training and validation performance, so the structural signal is not an artifact of overfitting.

Conditional sign reversals. Two predictors carry the opposite sign from their simple correlations with the outcome. Population density and crowded housing both correlate positively with the shooting rate on their own but turn protective once we hold the vulnerability profile constant – conditional on disadvantage, denser and more crowded tracts have lower rates. Limited-English share is also protective, consistent with the low violence of immigrant-enclave neighborhoods despite material hardship, though it was already weakly negative bivariate and so is not itself a reversal. Collinearity in the final model is mild (all variance-inflation factors below 3.5), so we read these as genuine conditional effects rather than artifacts.

Model fit and assumption checks. We built the candidate pool so that no SVI composite entered alongside its own components, and the overlapping education and CDC measures dropped during selection, leaving all variance-inflation factors mild (maximum 3.45). The Breusch-Pagan test rejects constant variance ($p = 4 \times 10^{-13}$) and Shapiro-Wilk rejects

residual normality ($p = 9 \times 10^{-5}$); the departure is a heavy upper tail, with the most violent tracts under-predicted even after the log transform. We handle these violations in the inference rather than ignore them – all tests use HC3 robust standard errors, and with $n = 1,680$ the central limit theorem additionally protects the coefficient tests. We do not report prediction intervals, which would not be trustworthy under these tails.

Poisson / Negative Binomial Model

Methods Shooting counts are non-negative, discrete events that are typically right-skewed, with many tracts experiencing few incidents and a smaller number of tracts having substantially higher counts. This distributional structure motivates the use of count regression models. The Poisson GLM is the canonical approach for such data, assuming the conditional mean and variance are equal. However, preliminary checks on the training data revealed substantial overdispersion (Pearson dispersion statistic = 7.52), indicating that the variance exceeds the mean. To avoid the underestimated standard errors and inflated statistical significance that result from ignoring this overdispersion, we implement a Negative Binomial GLM, which introduces a dispersion parameter (α) to model the extra variance.

To ensure robust inference, we screen for influential tracts using Cook’s distance calculated from an initial Negative Binomial fit on the training data. Tracts exceeding the conventional $4/n$ threshold are removed from the training set only; this flagged 100 of the 1,311 training tracts, leaving 1,211 training observations for subsequent model fitting. Following outlier removal, we perform variable selection using a bidirectional stepwise procedure that minimizes validation-set Poisson deviance. This validation-driven approach prefers predictors that improve out-of-sample predictive performance, balancing model fit against overfitting. As a robustness check, we also fit an L1-regularized (LASSO) Poisson GLM, with the regularization strength selected by minimizing validation-set deviance. This comparison allows us to distinguish between predictors that are robustly informative across methods and those that are method-dependent.

After variable selection, the chosen model is refit on the combined training and validation data ($n = 1,648$) using the Negative Binomial family with the estimated dispersion parameter. All reported inferential results (coefficients, standard errors, p-values) are based on this final refit. Final predictive performance is assessed on the held-out test set ($n = 437$) using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), reported on the original count scale.

Results The count model reveals several key insights into the structural determinants of shooting violence across New York City census tracts.

Overdispersion and model choice. The initial Poisson model fit on the training data produced a Pearson dispersion statistic of 7.52, confirming substantial overdispersion. The

final Negative Binomial model estimated a dispersion parameter of $\hat{\alpha} = 0.408$, further validating the choice of this family. Critically, the Negative Binomial standard errors are systematically larger than those from an uncorrected Poisson model, with inflation factors ranging from 1.65 to 2.56 across predictors. This underscores that a naive Poisson model would considerably overstate the statistical significance of many predictors: under the uncorrected Poisson fit, all 8 of 8 retained predictors are significant at $p < 0.05$, but only 6 of 8 remain significant once the Negative Binomial correction is applied — `cdc_pct_crowded_housing` ($p = 0.522$) and `cdc_pct_disabled` ($p = 0.162$) are no longer statistically distinguishable from zero. The substantially lower AIC for the Negative Binomial model (8,468.2) compared with the Poisson model (11,749.6), both evaluated on the same combined training-validation data, is consistent with the finding of substantial overdispersion in the count data that the Poisson model fails to capture – the AIC gap reflects how much of that overdispersion the added dispersion parameter is accounting for, rather than a generic claim that the Negative Binomial model fits the data better in absolute terms.

Figure 2 shows this overdispersion directly rather than through a single summary statistic. Training tracts are grouped into bins of similar predicted mean shooting count, and the observed variance within each bin is plotted against that mean. Under the Poisson assumption, variance should equal the mean (dashed line); instead, observed variance grows roughly quadratically, tracking the fitted Negative Binomial curve $\text{Var} = \mu + \hat{\alpha}\mu^2$ (using the training-set dispersion estimate, $\hat{\alpha} = 0.662$) far more closely than the Poisson line across most of the range. The fit is close for the majority of tracts, which have predicted means below roughly 15–20 shootings; at the highest predicted counts, observed variance falls somewhat below the fitted curve, consistent with the small number of high-shooting tracts that later drive most of the residual deviance discussed below.

Variable selection and multivariable associations. The validation-driven stepwise procedure retained 8 of the 17 candidate predictors as providing the most incremental predictive value: percentage Black, percentage Hispanic, median household income, percentage with a bachelor’s degree or higher, and four CDC SVI indicators — disabled, limited English proficiency, crowded housing, and no vehicle access. As a robustness check, the LASSO-regularized Poisson model selected a larger set of 12 predictors. The intersection of the two methods provides a set of seven core predictors — percentage Black, percentage Hispanic, no vehicle access, median household income, crowded housing, disabled, and limited English proficiency — that appear to be robustly associated with shooting rates regardless of the selection method.

The coefficient estimates, when exponentiated, yield incidence rate ratios (IRRs) that are directly interpretable as multiplicative changes in the expected shooting rate per standard-deviation increase in the predictor. The strongest positive as-

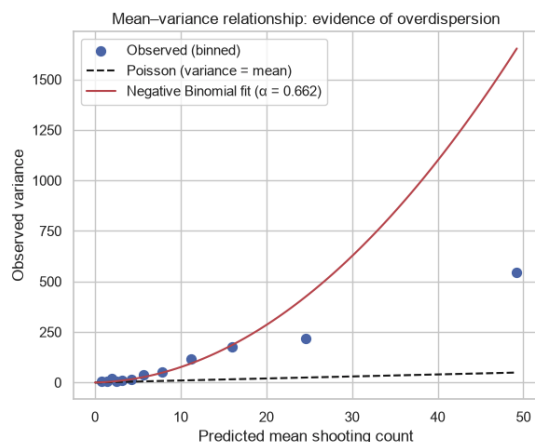


Figure 2: Mean–variance relationship among training-set census tracts, binned by predicted mean shooting count. The dashed line shows the Poisson assumption (variance = mean); the solid line shows the fitted Negative Binomial relationship using the training-set dispersion estimate ($\hat{\alpha} = 0.662$). Observed variance (blue points) tracks the Negative Binomial curve far more closely than the Poisson line across most of the range, visually confirming the overdispersion detected by the Pearson dispersion statistic (7.52). The two highest-count bins fall below the fitted curve, foreshadowing the localized misspecification discussed later in this section.

sociation is observed for the percentage of Black residents (IRR = 2.55, $p < 0.001$), followed by the percentage of Hispanic residents (IRR = 1.54, $p < 0.001$) and the percentage without vehicle access (IRR = 1.41, $p < 0.001$). Median household income (IRR = 0.83, $p < 0.001$) and the percentage with a bachelor’s degree or higher (IRR = 0.90, $p = 0.027$) are protective, consistent with their roles elsewhere in the analysis. Percentage with limited English proficiency is also protective but more modestly than in earlier specifications (IRR = 0.85, $p < 0.001$). Percentage disabled and percentage in crowded housing are not statistically distinguishable from zero once overdispersion is accounted for.

The racial-composition coefficients are large and should be read the same way the Logistic and Linear models’ race/ethnicity coefficients are read elsewhere in this report: as the association that remains after conditioning on income, education, and the CDC vulnerability measures included here, embedded in this city’s specific geography of segregation and disinvestment, not as an independent or causal effect of race itself (Krivo & Peterson, 1996; Lee et al., 2024; Sampson & Groves, 1989).

A particularly instructive case is the poverty rate. In a standalone bivariate Negative Binomial model, a one-standard-deviation increase in poverty is associated with an estimated 86% increase in the expected shooting rate (IRR ≈ 1.86 , $p < 0.001$). This strong marginal association aligns with the foundational literature on concentrated disadvantage (Krivo & Pe-

tersen, 1996; Sampson & Groves, 1989). However, poverty was not retained in the final stepwise multivariable model, though it was retained by the LASSO robustness check, reflecting its collinearity with other retained SVI and ACS indicators. A variance-inflation-factor (VIF) screen of the 17 standardized candidate predictors did not, on its own, flag severe multicollinearity: the highest values were concentrated among the three education-attainment measures (percentage without a high school diploma = 7.30, percentage with a high school education or higher = 7.28, percentage with a bachelor’s degree or higher = 7.09), with every predictor, including the poverty rate itself (VIF = 2.94), below the conventional severity threshold of 10. This illustrates why a VIF cutoff alone is an incomplete safeguard here: pairwise and aggregate VIF can look acceptable even as a predictor’s incremental contribution is fully absorbed by a correlated set of other retained variables, which is exactly what the validation-driven stepwise procedure – rather than a VIF threshold – is designed to detect. This result does not imply that poverty is unimportant; rather, it indicates that its effect on shooting rates is mediated by or confounded with these other dimensions of structural disadvantage, a nuance that reinforces the importance of multidimensional vulnerability frameworks (McMillan et al., 2024).

Model fit: a cautionary note. Although the deviance-to-degrees-of-freedom ratio for the final model is 1.120 – close to the ideal value of 1 for a well-specified Negative Binomial model – the formal deviance goodness-of-fit test rejects adequate fit ($p = 0.0005$). This apparent tension indicates that while the dispersion correction resolves the aggregate mean-variance mismatch, localized misspecification remains: a small number of tracts contribute disproportionately to the residual deviance. This foreshadows the geographic concentration of large residuals documented below.

Geographic patterns in model residuals. As a final diagnostic, we examine the spatial distribution of the largest model residuals. Among the top 20 tracts by squared Pearson residual, Queens accounts for 45% (9 of 20) and Brooklyn accounts for 40% (8 of 20), compared with baseline shares of approximately 31% and 35% of all census tracts, respectively; Manhattan and Staten Island contribute the remaining 2 and 1 tracts. Queens is therefore disproportionately represented relative to its overall share of tracts, while Brooklyn’s share is only modestly above baseline. This descriptive pattern suggests that the model does not fully capture localized neighborhood heterogeneity that is not accounted for by the socioeconomic and SVI predictors. Potential explanations include unmeasured factors such as localized policing practices, drug market activity, gang territories, or physical infrastructure (e.g., transportation hubs and commercial corridors) that have been shown to concentrate violence (Ceccato & Westman, 2024).

Logistic Regression Model

Methods Logistic regression was used to classify census tracts according to whether they experienced relatively high shooting rates. The binary outcome as high-shooting-tract, was defined as 1 when a tract's shooting rate per 1,000 residents exceeded the citywide median and 0 otherwise. Defining the outcome using the population-adjusted shooting rate rather than the raw shooting count accounts for differences in tract population size.

The model considered the same 17 standardized candidate predictors representing demographic, socioeconomic, educational, employment, and CDC Social Vulnerability Index characteristics. To prevent target leakage, shooting count, shooting rate, and population were excluded from the predictor set because the classification outcome was derived from the population-adjusted shooting rate. Influential observations were screened on the training data using Cook's distance with the conventional $4/n$ threshold, while the validation and test sets remained unchanged.

Variable selection was performed using bidirectional stepwise selection, with validation-set log-loss as the selection criterion. The classification probability threshold was also tuned using the validation set rather than automatically using the conventional 0.50 cutoff. After variable and threshold selection, the selected logistic model was refit using the combined training and validation data, while the held-out test set was used only once for final evaluation. Model performance was evaluated using AUC, log-loss, accuracy, precision, recall, and F1 score. Variance-inflation factors were examined to assess multicollinearity among the retained predictors. All analyses used a fixed random seed to ensure reproducibility.

Results The logistic model reframes the analysis as classification, asking how reliably a census tract's high-shooting status can be identified from its socioeconomic and vulnerability profile alone.

Gun violence is sharply concentrated in space. The single most salient feature of the data is spatial concentration. Figure 1 maps the tract-level shooting rate across the city. A large majority of tracts experience little or no gun violence, while a minority, clustered in the South Bronx, central and northern Brooklyn, and northern Manhattan, carry rates an order of magnitude higher than the citywide norm. This is the empirical signature that motivates a neighborhood-level analysis in the first place, and it echoes the micro-geography of violence documented in other cities (Baller et al., 2001; Braga et al., 2010). Substantively, it implies that violence-reduction resources allocated uniformly across the city would be poorly targeted, and that the practical value of a statistical model lies precisely in its ability to distinguish the high-risk minority of tracts from the low-risk majority.

The logistic model fits well and agree on the socioeconomic signal. The logistic model is the strongest performer for its

task, achieving a test AUC of 0.89 with high precision (0.85) and moderate recall (0.75) at the validation-chosen threshold.

A suppression effect for limited English. The limited-English result runs counter to its raw bivariate correlation with violence and is therefore more likely to reflect a genuine conditional relationship. Limited-English households are concentrated in immigrant enclaves that may possess protective social structures not captured by our other covariates, a pattern consistent with the "immigrant paradox" documented in urban crime research. The distinction matters: both results look surprising in a coefficient table, but they arise from different statistical mechanisms and warrant different substantive interpretations.

High-violence tracts are highly separable. The classification model reframes the analysis in terms of practical risk identification, and its results are summarized in Table 1 and Figure 3. The educational-attainment share carries the single largest odds ratio: a one-standard-deviation increase in the bachelor's-degree share is associated with an 85% reduction in the odds of a tract being high-shooting. Car-lessness, limited-English share, and single-parent households follow with substantial effects, and disability and male share contribute smaller but still significant effects. This model is comparatively well conditioned: variance-inflation factors among its nine retained predictors top out at 4.0 (household income), well under the conventional concern threshold of 10, so unlike the rate model these odds ratios can be read with reasonable confidence in both their direction and their approximate magnitude, not merely their direction. Household income, housing cost burden, and crowded housing do not reach significance in this specification, though all three point in the expected (protective or risk, respectively) direction. Figure 3A shows the receiver operating characteristic on the held-out test set: with an AUC of 0.89, the model separates high-from low-shooting tracts far better than chance, and the confusion matrix at the validation-chosen threshold (Figure 3B) shows high precision (0.85) with moderate recall (0.75): the tracts it flags are usually genuinely high-shooting, at the cost of missing some. A calibration check on the test set (predicted probabilities binned and compared against observed frequencies) tracks the diagonal closely, so the model's probabilities are trustworthy on their own scale and not just for ranking tracts. This directly answers one of our motivating questions in the affirmative: high-shooting tracts can be classified with high accuracy from socioeconomic and vulnerability indicators alone, without any information about past violence.

The models satisfy their diagnostic checks. Collinearity is moderate rather than severe in both retained-predictor sets: the logistic model's nine retained predictors top out at 4.0 (household income), comfortably under the conventional concern threshold of 10. The logistic model's McFadden

Table 1 below shows Predictors are standardized, so each odds ratio is the multiplicative change in odds per one-standard deviation increase; ordered by effect magnitude with 95% confidence intervals. The sign on percent limited English runs counter to its raw bivariate correlation with violence. Variance inflation factors among the nine predictors do not exceed 4.0, in which collinearity is only a modest concern.

Table 1: Logistic regression odds ratios for classifying a tract as high-shooting (above the citywide median shooting rate).

Predictor (per 1 SD)	Odds Ratio	95% CI
% bachelor's or higher	0.15	[0.11, 0.20]***
% No Vehicle	4.06	[3.27, 5.05]***
% limited English	0.28	[0.23, 0.35]***
% single parent	2.92	[2.29, 3.72]***
% disabled	1.333	[1.11, 1.59]**
Housing cost burden	0.80	[0.61, 1.05]
% crowded housing	0.84	[0.69, 1.02]
% male	0.84	[0.72, 0.98]*
Median household income	0.87	[0.65, 1.16]

Predictors are standardized with significance: * $p < .05$, ** $p < .01$, *** $p < .001$.

pseudo- R^2 of 0.45 and likelihood-ratio deviance test indicated a substantial improvement over an intercept-only baseline.

Answering the research questions. The poverty rate itself does not survive multivariable selection in any of the three models, but its own bivariate association with every outcome is overwhelming: fit alone (train+validation, same family as each final model), a one-SD increase in poverty rate corresponds to an odds ratio of 3.08 for classifying a tract as high-shooting ($p < 10^{-50}$). This direct check confirms poverty is exactly as strong a correlate of tract-level violence as the literature predicts, even though it drops out of the parsimonious multivariable models. Does educational attainment remain associated with lower violence after controlling for income and demographic composition? Largely yes: the bachelor's-degree share is protective and significant in the classification model. Does the CDC SVI improve prediction beyond standard ACS variables? Our answer is a qualified yes: the SVI measures carry clear predictive signal and several are among the strongest coefficients, but they are correlated with one another and with the ACS variables, so variable selection retains only a compact subset in each model. And can high-shooting tracts be classified from these indicators? Emphatically yes, with a held-out AUC of 0.89.

The above Fig 3(A) shows held-out test performance of the logistic regression model. This ROC curve demonstrates strong classification performance (AUC = 0.888), high shooting tracts are highly separable from the rest of the city using socioeconomic and vulnerability indicators alone. Fig 3(B) above shows confusion matrix with 180 true negatives, 169 true positives, 31 false positives, and 57 false negatives, indi-

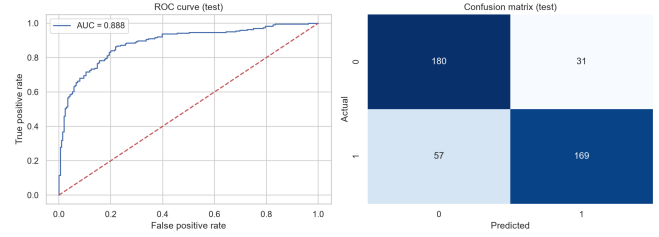


Figure 3: Fig 3(A) ROC curve and Fig 3(B) Confusion matrix

cating that the model correctly distinguishes most above- and below-median shooting tracts

Comparative Summary of Models

Across all three frameworks, two associations prove remarkably consistent. Minority and vulnerability composition is the dominant risk signal in both the linear and Negative Binomial models, though each uses it differently: the SVI minority-status ranking is by far the largest standardized risk factor in the linear model (+0.38 per SD, essentially matched only by no-vehicle access), while percentage Black is the single largest predictor in the Negative Binomial model (IRR = 2.55, $p < 0.001$). As discussed in each model's own results, this should be read as the association that remains after conditioning on income, education, and vulnerability measures – reflecting this city's specific geography of segregation and disinvestment, not an independent or causal effect of race itself (Krivo & Peterson, 1996; Lee et al., 2024; Sampson & Groves, 1989).

A second, more modest pattern also replicates: limited English proficiency is associated with a lower shooting rate after adjusting for other predictors in all three models. It appears as a negative standardized coefficient (−0.30 per SD, $p < 0.001$) in the linear model, a below-one incidence rate ratio (IRR = 0.85, $p < 0.001$) in the Negative Binomial model, and a below-one odds ratio in the logistic model. Replication across three different outcome definitions – continuous rate, count, and median-threshold classification – and three estimation methods makes this a genuine cross-model pattern, though it should be read as directional agreement rather than a precisely sized effect: the Negative Binomial effect is considerably weaker than the linear and logistic effects suggest, and the logistic model's own diagnostics attribute part of its coefficient's swing to multicollinearity.

The linear model contributes two points the other two cannot. First, it recovers a direct, significant poverty effect: whereas poverty dropped out of the Negative Binomial stepwise fit through collinearity (though it was retained by the LASSO robustness check), it survives selection in the linear model and is positive and significant (+0.12 per SD, $p < 0.001$), giving a multivariable estimate of the poverty–violence association that the count model could recover only bivariate (IRR ≈ 1.86). Second, it is the least affected by multicollinearity of the three – every variance-inflation factor

stays below 3.5, against higher values in the logistic model. It captures minority composition through the single SVI theme ranking rather than overlapping race and vulnerability components, so its coefficients can be read at close to face value.

Turning to the count model, the intersection of the Negative Binomial model’s stepwise and LASSO variable selection identified seven core predictors – percentage Black, percentage Hispanic, no-vehicle access, median household income, crowded housing, disabled, and limited English proficiency – of which percentage Black and no-vehicle access show the largest effect sizes (IRR = 2.55 and 1.41, respectively). No-vehicle access is a consistent cross-model risk factor: it is a strong positive predictor in both the linear model (+0.31 per SD) and the count model.

The Negative Binomial comparison also revealed a trade-off internal to the count-modeling approach itself: an uncorrected Poisson fit on the same 8 predictors achieved a lower test RMSE (10.918 vs. 11.708 for the Negative Binomial model) but understated standard errors by a factor of 1.65–2.56, illustrating that predictive accuracy and inferential reliability are not the same objective.

Summary of Model Performance

To provide a direct comparison of predictive performance across the three frameworks, we evaluate each model on its held-out test set, using the metrics appropriate to its outcome. The three models target different outcomes, so these figures are not directly comparable across models: the Multiple Linear model reports RMSE and MAE on the rate-per-1,000 scale (back-transformed from the fitted $\log(1 + \text{rate})$; test $R^2 = 0.35$), the Negative Binomial model on the raw count scale, and the Logistic model by AUC and classification accuracy. Table 2 summarizes the key metrics.

Table 2: Held-out Test Set Performance For The Three Models.

Model	Test RMSE / AUC	MAE / Accuracy
Multiple Linear	3.06	1.56
Negative Binomial	11.708	6.240
Logistic	0.888874	0.799792

Conclusions

Summary of Findings

This study investigates the relationship between neighborhood socioeconomic vulnerability and shooting violence across New York City using three complementary regression frameworks: Multiple Linear, Negative Binomial (Poisson), and Logistic regression.

In the Negative Binomial model, the percentage of Black residents and the percentage of Hispanic residents emerged as the strongest positive predictors, alongside a lack of vehicle access, while median household income, educational attainment, and limited English proficiency showed protective associations. As with the Logistic and Linear models,

the racial-composition coefficients should be read as residual associations after conditioning on income, education, and vulnerability measures, embedded in this city’s geography of segregation and disinvestment, rather than as independent or causal effects of race itself. Poverty, while strongly predictive in bivariate analysis (IRR ≈ 1.86), was not retained in the final stepwise multivariable model due to collinearity with other vulnerability indicators, highlighting that the effect of poverty on violence is mediated by or confounded with its correlated social and economic conditions. The model comparison also revealed a genuine trade-off between predictive accuracy and inferential reliability: the uncorrected Poisson model achieved the lowest test RMSE (10.918 vs. 11.708 for the Negative Binomial model), but its standard errors were understated by a factor of 1.65–2.56, overstating the significance of all 8 predictors where only 6 of 8 hold up under the dispersion-corrected model — making the Negative Binomial model the superior choice for drawing reliable statistical conclusions, even though it is not the single best predictor of raw counts. Relatedly, although the final model’s deviance-to-df ratio (1.120) suggests reasonable global fit, the formal goodness-of-fit test still rejects adequate fit ($p = 0.0005$), consistent with the geographic clustering of large residuals in Queens and Brooklyn described below – global fit statistics can look acceptable while masking substantial localized misspecification.

The Logistic model, classifying tracts above and below the citywide median shooting rate, achieved the strongest discriminative performance of the three frameworks (test AUC = 0.888, accuracy = 0.799). Educational composition, insurance status, lack of vehicle access, household structure, and unemployment were its strongest discriminators. As with the Negative Binomial model, several adjusted associations in the Logistic model diverge sharply from their raw bivariate direction. The model diagnostics indicate that strong multicollinearity among the retained predictors should be considered when interpreting unusually large adjusted odds ratios. One finding does replicate across both completed models despite their different outcome definitions: limited English proficiency is associated with lower shooting risk after adjustment, appearing as a protective association in both the Negative Binomial and Logistic models. This directional agreement across two different estimation approaches strengthens the case that it reflects a real, if not precisely sized, pattern rather than noise in a single model.

In the Multiple Linear Regression model, fit to the continuous shooting rate on the log scale, the SVI minority-status ranking was the strongest predictor (+0.38 per SD), followed by no-vehicle access (+0.31) and the poverty rate (+0.12), with educational attainment, population density, and limited-English share protective. Under HC3 robust standard errors all ten retained predictors were significant, and with every variance-inflation factor below 3.5 the coefficients can be read at face value. Notably, the linear model recovered a direct, significant poverty effect (+0.12 per SD, $p < 0.001$) that dropped

out of the count model's stepwise selection. It explained 58% of out-of-sample variation on the modeled log scale ($R^2=0.58$) and 35% on the back-transformed rate scale, and its heavy-tailed residuals – the most violent tracts under-predicted – reinforce the same message the count model's Queens and Brooklyn residuals send: socioeconomic structure explains much, but not all, of the geography of shooting violence.

The geographic concentration of large residuals in Queens and Brooklyn further demonstrates that socioeconomic predictors alone do not fully capture the heterogeneity of shooting violence. This aligns with our prior expectations that violence would remain geographically clustered even after accounting for observed covariates, reflecting persistent structural inequality and localized social dynamics that are difficult to measure with aggregate census data. Taken together, the findings affirm the literature on social determinants of firearm violence while offering a nuanced view that distinguishes between marginal associations, multivariable predictive utility, and the robustness of inference under different modeling assumptions.

Limitations and Future Work

Despite the integrative design of this study, several limitations should be acknowledged. First, the analysis relies on cross-sectional data; we cannot establish causal relationships between socioeconomic conditions and shooting rates, nor can we fully disentangle the reinforcing cycles of disadvantage and violence documented in longitudinal studies (Semenza et al., 2023). The observed associations are correlational and should be interpreted within the context of social determinants. Relatedly, associations estimated at the tract level need not hold at the individual level (the ecological fallacy), and the census tract itself is an administrative unit whose boundaries are somewhat arbitrary, so our findings are subject to the modifiable areal unit problem and could shift under a different spatial aggregation.

Second, while the CDC SVI provides a multidimensional measure of vulnerability, it is not exhaustive. Key structural factors such as local policing intensity, the density of alcohol outlets, the availability of green space, and public transit accessibility were not included in our predictor set, and their omission likely contributes to the residual geographic clustering observed in Brooklyn (Ceccato & Westman, 2024). Future work should integrate these environmental and institutional covariates, potentially using spatial regression techniques such as Spatial Error Models or Geographically Weighted Regression to account for spatial dependence.

Third, the NYPD shooting data captures only reported incidents, which may be subject to reporting biases across neighborhoods, and our reliance on census tracts as the unit of analysis imposes a static geographic boundary that may not reflect the actual social networks and activity spaces in which violence occurs. Future research could explore alternative spatial units, such as block groups or micro-place grids, and incorpo-

rate social network data to address the contagion mechanisms identified in the literature (Papachristos, 2009).

Fourth, our variable selection procedures (stepwise and LASSO) are designed to optimize predictive performance on the validation set, but they may overfit to the specific data split and the characteristics of NYC. The generalizability of the selected predictors to other metropolitan areas with different demographic and structural contexts remains an open question. A replication of this multi-model framework in cities such as Chicago, Los Angeles, or Philadelphia would provide a stronger empirical basis for identifying universal versus city-specific determinants of firearm violence.

Fifth, the SVI and ACS predictors measure overlapping dimensions of disadvantage, so moderate multicollinearity remains: several candidate predictors (poverty among them) are absorbed by correlated retained variables during selection, and a few coefficients that reverse their raw bivariate sign – such as limited-English proficiency – are best read as conditional suppression effects rather than isolated causal magnitudes. Regularized regression – ridge or elastic-net penalization – or substituting the SVI's published theme indices for their overlapping individual components (as the linear model does) would likely yield more stable effect estimates and is a natural next step for any of the three models.

Finally, the model comparison revealed the persistent tension between prediction and inference. The competitive performance of the LASSO-regularized model suggests that future work should consider ensemble methods or regularization techniques that balance bias and variance while retaining interpretability. Extending this analysis to incorporate machine learning methods – such as Random Forests or Gradient Boosting – could improve predictive accuracy, though at the cost of coefficient-level interpretability. For policy-oriented research, maintaining interpretability is essential for translating findings into actionable, place-based prevention strategies.

In summary, this study provides robust evidence that structural socioeconomic vulnerability is a consistent and multifaceted predictor of shooting violence in New York City. By systematically comparing three complementary regression approaches, we identify which dimensions of disadvantage are most consistently associated with firearm outcomes and highlight the importance of accounting for overdispersion and spatial heterogeneity. These findings support a shift from deficit-based models to comprehensive, place-based interventions that simultaneously address economic resources, housing stability, education, and community cohesion – the very structural drivers that this analysis has shown to be inextricably linked to gun violence.

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Team Member Contributions

All five team members contributed equally and collaboratively throughout every stage of the project. Each member regularly attended team meetings, participated in discussions and decision-making, shared ideas, reviewed the work of other members, and contributed to both the analytical and written components of the project. The contributions described below reflect the shared responsibilities of all team members.

Leroy Bagley

Co-led the MLR analysis and contributed equally across the project, including topic selection, dataset identification and acquisition, literature review and source tracking, analysis-plan development, and all major data-engineering steps (loading, cleaning, merging, spatial joining, and EDA). Performed the full Multiple Linear regression modeling workflow in R and generated the accompanying analysis write-up. Contributed written sections covering methodology, data preparation, exploratory analysis, modeling, and results, and authored R code for data processing, spatial operations, model implementation, diagnostics, and reproducible report preparation in Overleaf.

Gulmira Zhavgasheva

Co-led the MLR analysis and contributed equally across the project, including topic selection, dataset identification and acquisition, literature review and source tracking, analysis-plan development, and all major data-engineering steps (loading, cleaning, merging, spatial joining, and EDA). Performed the full Multiple Linear regression modeling workflow in R and generated the accompanying analysis write-up. Contributed written sections covering methodology, data preparation, exploratory analysis, modeling, and results, and authored R code for data processing, spatial operations, model implementation, diagnostics, and reproducible report preparation in Overleaf.

Henok Yohannes

Led the Logistic analysis and contributed equally across the project, including topic selection, dataset identification and acquisition, literature review and source tracking, analysis-plan development, and all major data-engineering steps (loading, cleaning, merging, spatial joining, and EDA). Performed the full logistic modeling workflow in R and generated the accompanying analysis write-up. Contributed written sections covering methodology, data preparation, exploratory analysis, modeling, and results, and authored R code for data processing, spatial operations, model implementation, diagnostics, and reproducible report preparation in Overleaf.

Evan Maurer

Co-led the Poisson analysis and contributed equally across the project, including topic selection, dataset identification and acquisition, literature review and source tracking, analysis-plan development, and all major data-engineering steps (loading, cleaning, merging, spatial joining, and EDA). Performed the full Poisson modeling workflow in Python using a Jupyter Notebook and produced the accompanying analysis write-up. Contributed written sections covering methodology, data preparation, exploratory analysis, modeling, and results, and authored R and Python code for data processing, spatial operations, model implementation, diagnostics, and reproducible report preparation in Overleaf.

Robert Arteaga

Co-led the Poisson analysis and contributed equally across the project, including topic selection, dataset identification and acquisition, literature review and source tracking, analysis-plan development, and all major data-engineering steps (loading, cleaning, merging, spatial joining, and EDA). Performed the full Poisson modeling workflow in Python using a Jupyter Notebook and produced the accompanying analysis write-up. Contributed written sections covering methodology, data preparation, exploratory analysis, modeling, and results, and authored R and Python code for data processing, spatial operations, model implementation, diagnostics, and reproducible report preparation in Overleaf.

Overall Contribution Statement

The project was completed as a collaborative team effort. Although individual members occasionally took the lead on particular tasks during different stages of the project, all five team members contributed equally to the overall project, from initial topic selection and data acquisition through literature review, analysis planning, data preparation, R and Python programming, EDA, statistical modeling, interpretation, report writing, referencing, Overleaf preparation, revisions, quality checks, and final submission. Major methodological and reporting decisions were discussed collectively, and all members participated in reviewing and refining the final work.